<u>Package Insert</u>

TELCORD H TABLETS
Telmisartan and Hydrochlorothiazide (40mg+12.5mg & 80mg+12.5mg)

Name and strength of active ingredient

TELCORD H 40/12.5
Telmisartan Ph. Eur. 40 mg
Hydrochlorothiazide Ph. Eur. 12.5 mg

TELCORD H 80+12.5
Telmisartan Ph. Eur. 80 mg
Hydrochlorothiazide Ph. Eur. 12.5 mg

Product Description

TELCORD H 40/12.5
White to off-white on one side and red on other side, biconvex, bilayer, oblong shaped, uncoated tablets debossed with 'T1' on red side and plain on other side, may have mottled appearance on red layer.

TELCORD H 80+12.5
White to off-white on one side and red on other side, biconvex, bilayer, oblong shaped, uncoated tablets debossed with 'T2' on red side and plain on other side, may have mottled appearance on red layer.

Pharmacological properties

Pharmacodynamic properties
Pharmacotherapeutic group: Angiotensin II antagonists and diuretics, ATC code: C09DA07

Mechanism of action
TELCORD H is a combination of an angiotensin II receptor antagonist, telmisartan, and a thiazide diuretic, hydrochlorothiazide. The combination of these ingredients has an additive antihypertensive effect, reducing blood pressure to a greater degree than either component alone.

TELCORD H once daily produces effective and smooth reductions in blood pressure across the therapeutic dose range.

<u>Telmisartan:</u>
Telmisartan is an orally effective and specific angiotensin II receptor (type AT1) antagonist. Telmisartan displaces angiotensin II with very high affinity from its binding site at the AT1 receptor subtype, which is responsible for the known actions of angiotensin II.

Telmisartan does not exhibit any partial agonist activity at the AT1 receptor. Telmisartan selectively binds the AT1 receptor. The binding is long lasting. Telmisartan does not show affinity for other receptors, including AT2 and other less characterised AT receptors. The functional role of these receptors is not known, nor is the effect of their possible overstimulation by angiotensin II, whose levels are increased by telmisartan. Plasma aldosterone levels are decreased by telmisartan. Telmisartan does not inhibit human plasma renin or block ion channels. Telmisartan does not inhibit angiotensin converting enzyme (kininase II), the enzyme which also degrades bradykinin. Therefore it is not expected to potentiate bradykinin-mediated adverse effects.

In man, an 80 mg dose of telmisartan almost completely inhibits the angiotensin II evoked blood pressure increase. The inhibitory effect is maintained over 24 hours and still measurable up to 48 hours.

<u>Hydrochlorothiazide:</u>

Hydrochlorothiazide is a thiazide diuretic. The mechanism of the antihypertensive effect of thiazide diuretics is not fully known. Thiazides effect the renal tubular mechanisms of electrolyte re-absorption, directly increasing excretion of sodium and chloride in approximately equivalent amounts. The diuretic action of hydrochlorothiazide reduces plasma volume, increases plasma renin activity, increases aldosterone secretion, with consequent increases in urinary potassium and bicarbonate loss, and decreases in serum potassium. Presumably through blockade of the renin-angiotensin-aldosterone system, co-administration of telmisartan tends to reverse the potassium loss associated with these diuretics.

With hydrochlorothiazide, onset of diuresis occurs in 2 hours, and peak effect occurs at about 4 hours, while the action persists for approximately 6 - 12 hours; the antihypertensive effect lasts for up to 24 hours.

Pharmacodynamics

<u>Telmisartan</u>
After the first dose of telmisartan, the antihypertensive activity gradually becomes evident within 3 hours. The maximum reduction in blood pressure is generally attained 4 weeks after the start of treatment and is sustained during long-term therapy.

The antihypertensive effect persists constantly over 24 hours after dosing and includes the last 4 hours before the next dose as shown by ambulatory blood pressure measurements. This is confirmed by through to peak ratios consistently above 80% seen after doses of 40 and 80 mg of telmisartan in placebo controlled clinical studies.

There is an apparent trend to a dose relationship to a time to recovery of baseline SBP. In this respect data concerning DBP are inconsistent.

In patients with hypertension telmisartan reduces both systolic and diastolic blood pressure without affecting pulse rate. The antihypertensive efficacy of telmisartan has been compared to antihypertensive drugs such as amlodipine, atenolol, enalapril, hydrochlorothiazide, losartan, lisinopril, ramipril and valsartan.

Upon abrupt cessation of treatment with telmisartan, blood pressure gradually returns to pre-treatment values over a period of several days without evidence of rebound hypertension. Telmisartan treatment has been shown in clinical trials to be associated with statistically significant reductions in Left Ventricular Mass and Left Ventricular Mass Index in patients with hypertension and Left Ventricular Hypertrophy.

Telmisartan treatment has been shown in clinical trials (including comparators like losartan, ramipril and valsartan) to be associated with statistically significant reductions in proteinuria (including microalbuminuria and macroalbuminuria) in patients with hypertension and diabetic nephropathy.

Hydrochlorothiazide

Hydrochlorothiazide inhibits sodium reabsorption primarily in the distal tubule, whereby a maximum of about 15% of sodium filtered by the glomerulus may be excreted. The extent of chloride excretion is roughly equivalent to that of sodium excretion. Potassium excretion is also increased by hydrochlorothiazide and is essentially determined by potassium secretion in the distal tubule and collecting duct (increased sodium and potassium ion exchange). Bicarbonate excretion may be increased by high hydrochlorothiazide doses as a result of carbonic anhydrase inhibition, resulting in urine alkalisation.

The saluretic/diuretic effect of hydrochlorothiazide is not significantly affected by acidosis or alkalosis.

Initially, the glomerular filtration rate is slightly reduced.

During long-term therapy with hydrochlorothiazide, calcium excretion via the kidneys is reduced, which may result in hypercalcaemia.

In hypertensive patients, hydrochlorothiazide has an antihypertensive effect. However, the mechanism has yet to be sufficiently clarified. It has been argued, for instance, that the effect of thiazide diuretics in terms of reducing vascular tone is due to a decrease in sodium concentration in the vascular wall and, hence, due to reduced responsiveness to norepinephrine.

In patients with chronic renal impairment (creatinine clearance less than 30 mL/min and/or serum creatinine above 1.8 mg/100 mL), hydrochlorothiazide has virtually no effect.

In patients with renal and ADH-sensitive diabetes insipidus, hydrochlorothiazide has an antidiuretic effect.

The incidence of dry cough was significantly lower in patients treated with telmisartan than in those given angiotensin converting enzyme inhibitors in clinical trials directly comparing the two antihypertensive treatments.

Clinical Trials

Prevention of cardiovascular morbidity and mortality

ONTARGET (ONgoin Telmisartan Alone and in Combination with Ramipril Global Endpoint Trial) compared the effects of telmisartan, ramipril and the combination of telmisartan and ramipril on cardiovascular outcomes in 25620 patients aged 55 years or older with a history of coronary artery disease, stroke, peripheral vascular disease, or diabetes mellitus accompanied by evidence of end-organ damage (e.g. retinopathy, left ventricular hypertrophy, macro- or microalbuminuria), which represents a broad cross-section of cardiovascular high risk patients.

Patients were randomized to one of the three following treatment groups: telmisartan 80 mg (n = 8542), ramipril 10 mg (n = 8576), or the combination of telmisartan 80 mg plus ramipril 10 mg (n = 8502), and followed for a mean observation time of 4.5 years. The population studied was 73% male, 74% Caucasian, 14% Asian and 43% were 65 years of age or older. Hypertension was present in nearly 83% of randomized patients: 69% of patients had a history of hypertension at randomization and an additional 14% had actual blood pressure readings above 140/90 mm Hg. At baseline, the total percentage of patients with a medical history of diabetes was 38% and an additional 3% presented with elevated fasting plasma glucose levels. Baseline therapy included acetylsalicylic acid (76%), statins (62%), beta-blockers (57%), calcium channel blockers (34%), nitrates (29%) and diuretics (28%).

The primary endpoint was a composite of cardiovascular death, non-fatal myocardial infarction, non-fatal stroke, or hospitalization for congestive heart failure.

Adherence to treatment was better for telmisartan than for ramipril or the combination of telmisartan and ramipril, although the study population had been pre-screened for tolerance to treatment with an ACE-inhibitor. The analysis of adverse events leading to permanent treatment discontinuation and of serious adverse events showed that cough and angioedema were less frequently reported in patients treated with telmisartan than in patients treated with ramipril, whereas hypotension was more frequently reported with telmisartan.

Telmisartan had similar efficacy to ramipril in reducing the primary endpoint. The incidence of the primary endpoint was similar in the telmisartan (16.7%), ramipril (16.5%) and telmisartan plus ramipril combination (16.3%) arms. The hazard ratio for telmisartan vs. ramipril was 1.01 (97.5% CI 0.93 - 1.10, p (non-inferiority) = 0.0019). The treatment effect was found to persist following corrections for differences in systolic blood pressure at baseline and over time. There was no difference in the primary endpoint based on age, gender, race, baseline therapies or underlying disease.

Telmisartan was also found to be similarly effective to ramipril in several pre-specified secondary endpoints, including a composite of cardiovascular death, non-fatal myocardial infarction, and non-fatal stroke, the primary endpoint in the reference study HOPE (The Heart Outcomes Prevention Evaluation Study), which had investigated the effect of ramipril vs. placebo. The hazard ratio of telmisartan vs. ramipril for this endpoint in ONTARGET was 0.99 (97.5% CI 0.90 - 1.08, p (non-inferiority) = 0.0004).

Combining telmisartan with ramipril did not add further benefit over ramipril or telmisartan alone. In addition, there was a significantly higher incidence of hyperkalaemia, renal failure, hypotension and syncope in the combination arm. Therefore the use of a combination of telmisartan and ramipril is not recommended in this population.

Epidemiological studies have shown that long-term treatment with hydrochlorothiazide reduces the risk of cardiovascular mortality and morbidity.

The effects of fixed dose combination of telmisartan/HCTZ on mortality and cardiovascular morbidity are currently unknown.

<u>Non-melanoma skin cancer</u>
Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High HCTZ use (≥50,000 mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.68-4.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to HCTZ: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use (~25,000 mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose (~100,000 mg).

Pharmacokinetic properties

Concomitant administration of hydrochlorothiazide and telmisartan has no effect on the pharmacokinetics of either drug.

<u>Absorption</u>
Telmisartan: Following oral administration peak concentrations of telmisartan are reached in 0.5 – 1.5 h after dosing. The absolute bioavailability of telmisartan at 40 mg and 160 mg was 42% and 58%, respectively. Food slightly reduces the bioavailability of telmisartan with a reduction in the area under the plasma concentration time curve (AUC) of about 6% with the 40 mg tablet and about 19% after a 160 mg dose. By 3 hours after administration plasma concentrations are similar whether telmisartan is taken fasting or with food. The small reduction in AUC is not expected to cause a reduction in the therapeutic efficacy.

The pharmacokinetics of orally administered telmisartan are non-linear over doses from 20 – 160 mg with greater than proportional increases of plasma concentrations (Cmax and AUC) with increasing doses. Telmisartan does not accumulate significantly in plasma on repeated administration.

Hydrochlorothiazide: Following oral administration of TELCORD H peak concentrations of hydrochlorothiazide are reached in approximately 1.0 – 3.0 hours after dosing. Based on cumulative renal excretion of hydrochlorothiazide the absolute bioavailability was about 60%.

Distribution

Telmisartan: Telmisartan is highly bound to plasma proteins (> 99.5%) mainly albumin and alpha1-acid glycoprotein. The apparent volume of distribution for telmisartan is approximately 500 litres indicating additional tissue binding.

Hydrochlorothiazide: Hydrochlorothiazide is 64% protein bound in the plasma and its apparent volume of distribution is 0.8 ± 0.3 l/kg.

Biotransformation and elimination

Telmisartan: Following either intravenous or oral administration of 14C-labelled telmisartan most of the administered dose (> 97%) was eliminated in faeces via biliary excretion. Only minute amounts were found in urine.

Telmisartan is metabolised by conjugation to form a pharmacologically inactive acylglucuronide. The glucuronide of the parent compound is the only metabolite that has been identified in humans.

After a single dose of 14C-labelled telmisartan the glucuronide represents approximately 11% of the measured radioactivity in plasma. The cytochrome P450 isoenzymes are not involved in the metabolism of telmisartan. Total plasma clearance of telmisartan after oral administration is > 1500 ml/min. Terminal elimination half-life was > 20 hours.

Hydrochlorothiazide: Hydrochlorothiazide is not metabolised in man and is excreted almost entirely as unchanged drug in urine. About 60% of the oral dose are eliminated as unchanged drug within 48 hours. Renal clearance is about 250 – 300 ml/min. The terminal elimination half-life of hydrochlorothiazide is 10 – 15 hours.

PK in specific populations

Elderly patients

Pharmacokinetics of telmisartan do not differ between the elderly and those younger than 65 years. In patients of more advanced age, one should be aware of possible renal impairment.

Gender

Plasma concentrations of telmisartan are generally 2 – 3 times higher in females than in males. In clinical trials however, no significant increases in blood pressure response or in the incidence of orthostatic hypotension were found in women. No dosage adjustment is necessary. There was a trend towards higher plasma concentrations of hydrochlorothiazide in female than in male subjects. This is not considered to be of clinical relevance.

Patients with renal impairment

Renal excretion does not contribute to the clearance of telmisartan. Based on modest experience in patients with mild to moderate renal impairment (creatinine clearance of 30 – 60 ml/min, mean about 50 ml/min) no dosage adjustment is necessary in patients with decreased renal function. Telmisartan is not removed from blood by haemodialysis. In patients with impaired renal function the rate of hydrochlorothiazide elimination is reduced.

In a typical study in patients with a mean creatinine clearance of 90 ml/min the elimination half-life of hydrochlorothiazide was increased. In functionally anephric patients the elimination half-life is about 34 hours.

Patients with hepatic impairment

Pharmacokinetic studies in patients with hepatic impairment showed an increase in absolute bioavailability up to nearly 100%. The elimination half-life is not changed in patients with hepatic impairment.

Indication/Usage

Treatment of essential hypertension.
As fixed dose combination TELCORD H is indicated in patients whose blood pressure is not adequately controlled on telmisartan alone.

Posology and method of administration

Adults

TELCORD H should be taken once daily. The dose of telmisartan could be up-titrated before switching to TELCORD H. Direct change from monotherapy to the fixed combinations may be considered.

- TELCORD H 40/12.5 mg may be administered in patients whose blood pressure is not adequately controlled by Telmisartan 40 mg or hydrochlorothiazide.
- TELCORD H 80/12.5 mg may be administered in patients whose blood pressure is not adequately controlled by Telmisartan 80 mg or by TELCORD H 40/12.5 mg.

Sodium or volume depletion should be corrected before treatment commencement with TELCORD H.

The maximum antihypertensive effect is generally attained with TELCORD H 4 – 8 weeks after the start of treatment.
When necessary, TELCORD H may be administered with another antihypertensive drug.

In patients with severe hypertension treatment with telmisartan at doses up to 160 mg alone and in combination with hydrochlorothiazide 12.5 - 25 mg daily was well tolerated and effective.

Special populations:

Geriatric patients

No dose adjustment is necessary for geriatric patients.

Paediatric patients

Safety and efficacy of TELCORD H have not been established in patients aged below 18 years. Use of TELCORD H is not recommended in children and adolescents.

Renal impairment

Due to the hydrochlorothiazide component, TELCORD H must not be used in patients with severe renal dysfunction (creatinine clearance < 30 ml/min). Loop diuretics are preferred to thiazides in this population. Experience in patients with mild to moderate renal impairment is modest but has not suggested adverse renal effects and dose adjustment is not considered necessary. Periodic monitoring of renal function is advised.

Telmisartan is not removed from blood by hemofiltration and is not dialyzable.

Hepatic impairment

In patients with mild to moderate hepatic impairment TELCORD H should be administered with caution. For telmisartan, the posology should not exceed 40mg once daily (see Contraindications). Thiazides should be used with caution in patients with impaired hepatic function.

Method of Administration:

TELCORD H tablets are for once-daily oral administration and should be swallowed whole with liquid. TELCORD H can be taken with or without food.

Handling Instructions

Due to the hygroscopic property of the tablets they should be taken out of the sealed blister shortly before administration.

Contraindication

- Hypersensitivity to the active ingredient, to any of the excipients, or to other sulphonamide-derived substances (hydrochlorothiazide is a sulphonamide-derived substance).
- Pregnancy
- Lactation
- Choleastasis and biliary obstructive disorders
- Severe hepatic impairment, coma hepatricum, hepatic precoma
- Severe renal impairment (creatinine clearance < 30 ml/min) or serum creatinine > 1.8 mg/100 ml), anuria, or acute glomerulonephritis
- Refractory hypokalaemia, hypercalcaemia
- Therapy-refractory hyponatraemia
- Hypovolaemia
- Symptomatic hyperuricaemia/ gout
- The concomitant use of TELCORD H with aliskiren is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 ml/min/1.73 m^2)

Special warnings and precautions for use

Hydrochlorothiazide

Acute Respiratory Toxicity

Very rare severe cases of acute respiratory toxicity, including acute respiratory distress syndrome (ARDS) have been reported after taking hydrochlorothiazide. Pulmonary oedema typically develops within minutes to hours after hydrochlorothiazide intake. At the onset, symptoms include dyspnoea, fever, pulmonary deterioration and hypotension. If diagnosis of ARDS is suspected, TELCORD H should be withdrawn and appropriate treatment given. Hydrochlorothiazide should not be administered to patients who previously experienced ARDS following hydrochlorothiazide intake.

Pregnancy
Angiotensin II receptor antagonists should not be initiated during pregnancy.
Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy.
When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and if appropriate, alternative therapy should be started.

Hepatic impairment
TELCORD H must not be given to patients with cholestasis, biliary obstructive disorders or severe hepatic insufficiency since telmisartan is mostly eliminated with the bile. These patients can be expected to have reduced hepatic clearance for telmisartan.
TELCORD H should be used with caution in patients with impaired hepatic function or progressive liver disease, since minor alterations of fluid and electrolyte balance may precipitate hepatic coma. There is no clinical experience with TELCORD H in patients with hepatic impairment.

Reno vascular hypertension
There is an increased risk of severe hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with medicinal products that affect the renin-angiotensin-aldosterone system.

Renal impairment and kidney transplantation
TELCORD H must not be used in patients with severe renal impairment (creatinine clearance < 30 ml/min).
There is no experience regarding the administration of TELCORD H in patients with severe renal impairment or with a recent kidney transplant. Experience with TELCORD H is modest in the patients with mild to moderate renal impairment, therefore periodic monitoring of potassium, creatinine and uric acid serum levels is recommended. Thiazide diuretic-associated azotaemia may occur in patients with impaired renal function.
Telmisartan is not removed from blood by hemofiltration and is not dialyzable.

Volume and/or sodium depleted patients
Symptomatic hypotension, especially after the first dose, may occur in patients who are volume and/or sodium depleted by vigorous diuretic therapy, dietary salt restriction, diarrhoea or vomiting. Such conditions, especially volume and/or sodium depletion, should be corrected before the administration of TELCORD H.

Isolated cases of hyponatraemia accompanied by neurological symptoms (nausea, progressive disorientation, apathy) have been observed with the use of HCTZ.

Dual blockade of the renin-Angiotensin-aldosterone system (RAAS)
As a consequence of inhibiting the renin-angiotensin-aldosterone system changes in renal function (including acute renal failure) have been reported in susceptible individuals, especially if combining medicinal products that affect this system. Dual blockade of the renin-angiotensin-aldosterone system (e.g. by adding an ACE-inhibitor or the direct renin-inhibitor aliskiren to an angiotensin II receptor antagonist) is not recommended and should therefore be limited to individually defined cases with close monitoring of renal function.

Other conditions with stimulation of the renin-Angiotensin-aldosterone system
In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with other medicinal products that affect this system has been associated with acute hypotension, hyperazotaemia, oliguria, or rarely acute renal failure.

Primary aldosteronism
Patients with primary aldosteronism generally will not respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of TELCORD H is not recommended.

Aortic and mitral valve stenosis, obstructive hypertrophic cardio myopathy
As with other vasodilators, special caution is indicated in patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.

Metabolic and endocrine effects
Thiazide therapy may impair glucose tolerance. In diabetic patients, dosage adjustments of insulin or oral hypoglycaemic agents may be required. Latent diabetes mellitus may become manifest during thiazide therapy.

An increase in cholesterol and triglyceride levels has been associated with thiazide diuretic therapy; however, at the 12.5 mg dose contained in TELCORD H, minimal or no effects were reported.

Hyperuricaemia may occur or frank gout may be precipitated in some patients receiving thiazide therapy.

Electrolyte imbalance
As for any patient receiving diuretic therapy, periodic determination of serum electrolytes should be performed at appropriate intervals.

Thiazides, including hydrochlorothiazide, can cause fluid or electrolyte imbalance (hypokalaemia, hyponatraemia, and hypochloraemic alkalosis). Warning signs of fluid or electrolyte imbalance are dryness of mouth, thirst, weakness, lethargy, drowsiness, restlessness, muscle pain or cramps, muscular fatigue, hypotension, oliguria, tachycardia, and gastro-intestinal disturbances such as nausea or vomiting.

Hypokalemia
Although hypokalaemia may develop with the use of thiazide diuretics, concurrent therapy with telmisartan may reduce diuretic-induced hypokalaemia. The risk of hypokalaemia is greatest in patients with cirrhosis of liver, in patients experiencing brisk diuresis, in patients who are receiving inadequate oral intake of electrolytes and in patients receiving concomitant therapy with corticosteroids or ACTH.

Hyperkalemia
Conversely, due to the antagonism of the angiotensin II (AT1) receptors by the telmisartan component of TELCORD H, hyperkalaemia might occur. Although clinically significant hyperkalaemia has not been documented with TELCORD H, risk factors for the development of hyperkalaemia include renal insufficiency and/or heart failure, and diabetes mellitus. Potassium-sparing diuretics, potassium supplements or potassium-containing salt substitutes should be co-administered cautiously with TELCORD H.

Hypercalcemia
Thiazides may decrease urinary calcium excretion and cause an intermittent and slight elevation of serum calcium in the absence of known disorders of calcium metabolism. Marked hypercalcaemia may be evidence of hidden hyperparathyroidism. Thiazides should be discontinued before carrying out tests for parathyroid function.

Hypomagnesaemia
Thiazides have been shown to increase the urinary excretion of magnesium, which may result in hypomagnesaemia.

Sorbitol
TELCORD H tablets 40/12.5 mg contains 169 mg sorbitol in each tablet. TELCORD H tablets 80/12.5 mg contains 338 mg sorbitol in each tablet.

Sorbitol is a source of fructose. TELCORD H tablets 80/12.5 mg is not recommended for use in patients with hereditary fructose intolerance (HFI).

Diabetes mellitus:
In diabetic patients with an additional cardiovascular risk, i.e. patients with diabetes mellitus and coexistent coronary artery disease (CAD), the risk of fatal myocardial infarction and unexpected cardiovascular death may be increased when treated with blood pressure lowering agents such as ARBs or ACE-inhibitors. In patients with diabetes mellitus CAD may be asymptomatic and therefore undiagnosed. Patients with diabetes mellitus should undergo appropriate diagnostic evaluation, e.g. exercise stress testing, to detect and to treat CAD accordingly before initiating treatment with TELCORD H.

Lactose:
The maximum recommended daily dose of TELCORD H contains 112 mg of lactose monohydrate in the dose strengths 40/12.5 mg and 80/12.5 mg.

Patients with the rare hereditary conditions of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Ischaemic heart disease:
As with any antihypertensive agent, excessive reduction of blood pressure in patients with ischaemic cardiopathy or ischaemic cardiovascular disease could result in a myocardial infarction or stroke.

General:
Hypersensitivity reactions to hydrochlorothiazide may occur in patients with or without a history of allergy or bronchial asthma, but are more likely in patients with such a history.

Exacerbation or activation of systemic lupus erythematosus has been reported with the use of thiazide diuretics.

Cases of photosensitivity reactions have been reported with thiazide diuretics (see Adverse reactions). If a photosensitivity reaction occurs during treatment, it is recommended to stop the treatment. If a re-administration of the diuretic is deemed necessary, it is recommended to protect exposed areas to the sun or to artificial UVA.

Choroidal Effusion, Acute Myopia and Secondary Angle-Closure Glaucoma:
Hydrochlorothiazide, a sulfonamide, can cause an idiosyncratic reaction, resulting in choroidal effusion with visual field defect, in acute transient myopia and acute angle-closure glaucoma. Symptoms include acute onset of decreased visual acuity or ocular pain and typically occur within hours to weeks of drug initiation. Untreated acute angle-closure glaucoma can lead to permanent vision loss. The primary treatment is to discontinue hydrochlorothiazide as rapidly as possible. Prompt medical or surgical treatments may need to be considered if the intraocular pressure remains uncontrolled. Risk factors for developing acute angle-closure glaucoma may include a history of sulfonamide or penicillin allergy.

Non-melanoma skin cancer
An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry (see Side Effects). Photosensitizing actions of hydrochlorothiazide could act as a possible mechanism for NMSC.

Patients taking hydrochlorothiazide should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions.
Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies.

Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. The use of hydrochlorothiazide may also need to be reconsidered in patients who have experienced previous NMSC.

Statement on Usage during Pregnancy, Lactation and Fertility

Pregnancy
Telmisartan:
The use of angiotensin II receptor antagonists is not recommended during the first trimester of pregnancy and should not be initiated during pregnancy. When pregnancy is diagnosed, treatment with angiotensin II receptor antagonists should be stopped immediately, and, if appropriate, alternative therapy should be started.

The use of angiotensin II receptor antagonists is contraindicated during the second and third trimester of pregnancy.

Non-clinical studies with telmisartan do not indicate teratogenic effect, but have shown fetotoxicity.

Angiotensin II receptor antagonists exposure during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia).

Unless continued angiotensin II receptor antagonist therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy.

Should exposure to angiotensin II receptor antagonists have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.
Infants whose mothers have taken angiotensin II receptor antagonists should be closely observed for hypotension.

Hydrochlorothiazide:
There is limited experience with hydrochlorothiazide during pregnancy, especially during the first trimester.

Hydrochlorothiazide crosses the placenta. Based on the pharmacological mechanism of action of hydrochlorothiazide its use during the second and third trimester may compromise foeto-placental perfusion and may cause foetal and neonatal effects like icterus, disturbance of electrolyte balance and thrombocytopenia.

Hydrochlorothiazide should not be used for gestational oedema, gestational hypertension or preeclampsia due to the risk of decreased plasma volume and placental hypoperfusion, without a beneficial effect on the course of the disease.

Hydrochlorothiazide should not be used for essential hypertension in pregnant women except in rare situations where no other treatment could be used.

Lactation

TELCORD H is contraindicated during lactation since it is not known whether telmisartan is excreted in human milk. Non-clinical studies have shown excretion of telmisartan in breast milk. Thiazides appear in human milk and may inhibit lactation.

Fertility

No studies on fertility in humans have been performed.
In non-clinical studies, an effect of telmisartan and hydrochlorothiazide on male and female fertility was not observed.

Effects on ability to drive and use machines

No studies on the effect on the ability to drive and use machines have been performed. However, when driving vehicles or operating machinery, it should be taken into account that dizziness, syncope or vertigo may occasionally occur when taking antihypertensive therapy.
If patients experience these adverse events, they should avoid potentially hazardous tasks such as driving or operating machinery.

Side effects/Adverse Reactions

Summary of the safety profile

The overall incidence of adverse events reported with TELCORD H was comparable to those reported with telmisartan alone in randomised controlled trials involving 1471 patients receiving telmisartan plus hydrochlorothiazide (835) or telmisartan alone (636). There was no dose-relationship to undesirable effects and there was no correlation with gender, age or race of the patients.

Tabulated summary of adverse reactions

Adverse reactions reported in clinical trials with telmisartan plus hydrochlorothiazide are shown below according to system organ class. Adverse reactions not observed in clinical trials with telmisartan plus hydrochlorothiazide but expected during treatment with TELCORD H based on the experience with telmisartan or hydrochlorothiazide alone are shown in the table below classified by MedDRA System organ class and MedDRA Preferred Terms.

MedDRA System Organ Class terminology	Adverse reactions
Infections and infestations	bronchitis[1]
	pharyngitis[1]
	sinusitis[1]
	upper respiratory tract infection[2]

	urinary tract infection[2]
	cystitis[2]
	sepsis (including fatal outcome)[2]
Neoplasms benign, malignant and unspecified (incl cysts and polys)	basal cell carcinoma[3]
	squamous cell carcinoma of skin[3]
	lip squamous cell carcinoma[3]
Blood and lymphatic system disorders	anaemia[2]
	thrombocytopenia[2]
	thrombocytopenic purpura[3]
	eosinophilia[2]
	aplastic anaemia[3]
	haemolytic anaemia[3]
	bone marrow failure[3]
	leukopenia[3]
	agranulocytosis[3]
Immun system disorders	lupus erythematosus*,1
	vasculitis necrotising3
	anaphylactic reaction2,3
	hypersensitivity2
Endocrine disorders	diabetes mellitus inadequatecontrol[3]
Metabolism and nutrition disorders	hypokalaemia[1]
	hyponatraemia[1]
	hyperuricaemia[1]
	hyperkalaemia[2]
	hypoglycaemia (in diabetic patients)[2]
	hypovolaemia[3]
	electrolyte imbalance[3]
	decreased appetite[3]
	hyperglycaemia[3]
	hyperlipidaemia[3]

	hypomagnesaemia[3]
	hypercalcaemia[3]
	hypochloraemic alkalosis[3]
	glycosuria[3]
Psychiatric disorders	anxiety[1]
	depression[1]
	restlessness[3]
Nervous system disorders	dizziness[1]
	syncope (faint)[1]
	paraesthesia[1]
	sleep disorder[1]
	insomnia[1]
	headache[3]
Eye disorders	visual impairment[1]
	vision blurred[1]
	angle-closure glaucoma[3]
	choroidal effusion[3]
Ear and labyrinth disorders	vertigo[1]
Cardiac disorders	arrhythmia[1]
	tachycardia[1]
	bradycardia[2]
Vascular disorders	hypotension[1]
	orthostatic hypotension[1]
Respiratory, thoracic and mediastinal disorders	dyspnoea[1]
	respiratory distress[1,3]
	pneumonitis[1,3]
	pulmonary oedema[1,3]
Uncommon	Cough
Very rare	Acute respiratory distress syndrome (ARDS)
	Interstitial lung disease

Gastrointestinal disorders	diarrhoea[1]
	dry mouth[1]
	flatulence[1]
	abdominal pain[1]
	constipation[1]
	dyspepsia[1]
	vomiting[1]
	gastritis[1]
	abdominal discomfort[2,3]
	pancreatitis[3]
	nausea[3]
Hepatobiliary disorders	abnormal hepatic function / liver disorder[§,1]
	jaundice (cholestatis intrahepatic)[3]
Skin and subcutaneous tissue disorders	angioedema (with fatal outcome)[1]
	erythema[1]
	pruritus[1]
	rash[1]
	hyperhidrosis[1]
	urticaria[1]
	eczema[2]
	drug eruption[2]
	toxic skin eruption[2]
	toxic epidermal necrolysis[3]
	lupus like syndrome (drug-induced cutaneous lupus erythematosus)[3]
	photosensitivity reaction[3]
	erythema multiforme[3]
Musculoskeletal and connective tissue disorders	back pain[1]
	muscle spasms (cramps in legs)[1]
	myalgia[1]

	arthralgia[1]
	pain in extremity (leg pain)[1]
	tendon pain (tendonitis like symptoms)[2]
Rare	Systemic lupus erythematosus
Renal and urinary disorders	renal impairment (including acute renal injury[1]),
Uncommon	Acute renal failure
Rare	Glucosuria
Reproductive system and breast disorders	erectile dysfunction[1]
General disorders and administration site conditions	chest pain[1]
	influenza-like illness[1]
	pain[1]
	asthenia (weakness)[2, 3]
	pyrexia[3]
Investigations	blood uric acid increased[1]
	blood creatinine increased[1]
	hepatic enzyme increased[1]
	blood creatine phosphokinase increased[1]
	haemoglobin decreased[2]

[1] Adverse reactions of FDC telmisartan + hydrochlorothiazide
[2] Adverse reactions of telmisartan as monotherapy in the indication hypertension or in patients 50 years or older at high risk of cardiovascular events
[3] Adverse reactions of hydrochlorothiazide as monotherapy
* based on post-marketing experience
§ Most cases of hepatic function abnormal / liver disorder from post-marketing experience with telmisartan occurred in patients in Japan, who are more likely to experience these adverse reactions

Neoplasms benign, malignant and unspecified (incl cysts and polyps)
Frequency ‘not known’: Non-melanoma skin cancer (Basal cell carcinoma and Squamous cell carcinoma)

Description of selected adverse reactions
Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between HCTZ and NMSC has been observed.

Drug Interactions

Interactions linked to telmisartan

Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with angiotensin converting enzyme inhibitors. Cases have also been reported with angiotensin II receptor antagonists, including telmisartan. Furthermore, renal clearance of lithium is reduced by thiazides so the risk of lithium toxicity could be increased with TELCORD H. Lithium and TELCORD H should only be co-administered under medical supervision and serum lithium level monitoring is advisable during concomitant use.
The potassium-depleting effect of hydrochlorothiazide is attenuated by the potassium-sparing effect of telmisartan. However, this effect of hydrochlorothiazide on serum potassium would be expected to be potentiated by other drugs associated with potassium loss and hypokalaemia (e.g. other kaliuretic diuretics, laxatives, corticosteroids, ACTH, amphotericin, carbenoxolone, penicillin G sodium, salicylic acid and derivatives).

If these drugs are to be prescribed with TELCORD H, monitoring of potassium plasma levels is advised.

Conversely, based on the experience with the use of other drugs that blunt the renin-angiotensin system, concomitant use of potassium-sparing diuretics, potassium supplements, salt substitutes containing potassium or other drugs that may increase serum potassium levels (e.g. heparin sodium) may lead to increases in serum potassium.
If these drugs are to be prescribed with TELCORD H, monitoring of potassium plasma levels is advised.

Periodic monitoring of serum potassium is recommended when TELCORD H is administered with drugs affected by serum potassium disturbances, e.g. digitalis glycosides, anti-arrhythmic agents and drugs known to induce torsades de pointes.

Treatment with non-steroidal anti-inflammatory drugs (NSAIDs) including ASA at anti-inflammatory dosage regimens, COX-2 inhibitors and non-selective NSAIDs is associated with the potential for acute renal insufficiency in patients who are dehydrated. Compounds acting on the Renin-Angiotensin-System like telmisartan may have synergistic effects. Patients receiving NSAIDs and telmisartan should be adequately hydrated and be monitored for renal function at the beginning of combined treatment.
The co-administration of NSAIDs may reduce the diuretic, natriuretic and antihypertensive effects of thiazide diuretics in some patients.

Telmisartan may increase the hypotensive effect of other antihypertensive agents.
Co-administration of telmisartan did not result in a clinically significant interaction with digoxin, warfarin, hydrochlorothiazide, glibenclamide, ibuprofen, paracetamol, simvastatin and

amlodipine. For digoxin a 20 % increase in median plasma digoxin trough concentration has been observed (39 % in a single case), monitoring of plasma digoxin levels should be considered.

In one study the co-administration of telmisartan and ramipril led to an increase of up to 2.5 fold in the AUC0-24 and Cmax of ramipril and ramiprilat. The clinical relevance of this observation is not known.

Interactions linked to hydrochlorothiazide (HCT)

The antihypertensive effect of HCT can be potentiated by other diuretics, antihypertensive agents, guanethidine, methyldopa, calcium antagonists, ACE inhibitors, ARBs, DRIs, beta-receptor blockers, nitrates, barbiturates, phenothiazines, tricyclic antidepressants, vasodilators or by alcohol consumption.

Salicylates and other non-steroidal anti-inflammatory drugs (e.g. indomethacin) may reduce the antihypertensive and diuretic effect of HCT. In patients taking high-dose salicylates, the toxic effect of salicylates on the central nervous system may be potentiated. In patients developing hypovolaemia during treatment with HCT, concomitant administration of non-steroidal anti-inflammatory drugs may trigger acute renal failure.
Co-administration of thiazides (including hydrochlorothiazide) and allopurinol may possibly increase the frequency of hypersensitivity reactions to allopurinol.

Co-administration of thiazides and amantadine may possibly increase the risk of amantadine-related adverse reactions.

There is an increased risk for the onset of hyperglycaemia with concomitant administration of HCT and beta-receptor blockers.

The effect of insulin or oral antidiabetics, uric acid-lowering agents, as well as norepinephrine and epinephrine, may be attenuated with concomitant use of HCT. An adjustment of the insulin or oral antidiabetic dosage may therefore be required.

In concomitant treatment with cardiac glycosides, it must be remembered that myocardial sensitivity to cardiac glycosides will be increased by any hypokalaemia and/or hypomagnesaemia that develops during HCT therapy, thereby potentiating the effects and adverse effects of these cardiac glycosides.

Concomitant use of HCT and kaliuretic diuretics (e.g. furosemide), glucocorticoids, ACTH, carbenoxolone, penicillin G, salicylates, amphotericin B, antiarrhythmics or laxatives may lead to increased potassium loss.
Concomitant use of natriuretic diuretics and antidepressants, antipsychotics or antiepileptics may lead to increased sodium loss.

Concomitant use of thiazide diuretics and cytotoxic agents (e.g. cyclophosphamide, fluorouracil, methotrexate) may lead to a reduction in the renal excretion of cytotoxic agents. Increased bone marrow toxicity (especially granulocytopenia) can be expected.

The bioavailability of thiazide diuretics may be increased by anticholinergic agents (e.g. atropine, biperiden). This is probably due to a decrease in gastrointestinal motility and the gastric emptying rate. In contrast, prokinetic medicinal products such as cisapride may reduce the bioavailability of thiazide diuretics.

Diuretics increase plasma lithium levels. As concomitant administration of HCT and lithium leads to potentiation of the cardio-and neurotoxic effects of lithium due to decreased lithium excretion, the lithium level must be monitored in patients receiving HCT and lithium. In patients in whom lithium has induced polyuria, diuretics can have a paradoxical antidiuretic effect.

The effect of curare-like muscle relaxants may be potentiated or prolonged by HCT. In cases where HCT cannot be discontinued before the use of curare-like muscle relaxants, the anaesthetist must be informed of the treatment with HCT.

Concomitant use of cholestyramine or colestipol reduces the absorption of HCT. However, the interaction may possibly be minimized by staggered dosing of hydrochlorothiazide and the resinate, so that hydrochlorothiazide is taken at least 4 hours before or 4-6 hours after administration of the resinate.
Concomitant use with vitamin D may reduce the excretion of calcium via the urine and potentiate the increase of calcium in serum.

When co-administered with calcium salts, hypercalcaemia may occur due to the increase in tubular calcium reuptake.

Concomitant use with ciclosporin may increase the risk of hyperuricaemia and gout-like complications.

Thiazides can increase the hyperglycaemic effect of diazoxide.

During concomitant use of methyldopa, there have been uncommon reports of haemolysis, caused by the formation of antibodies against hydrochlorothiazide.

Hydrochlorothiazide may reduce the response to adrenergic amines, such as norepinephrine.

Overdose
Symptoms
Limited information is available for TELCORD H with regard to overdose in humans.
The most prominent manifestations of telmisartan overdose were hypotension and tachycardia; bradycardia also occurred.

Overdose with hydrochlorothiazide is associated with electrolyte depletion (hypokalaemia, hypochloraemia) and dehydration resulting from excessive diuresis. The most common signs and symptoms of overdose are nausea and somnolence. Hypokalaemia may result in muscle spasm and/or accentuate cardiac arrhythmias associated with the concomitant use of digitalis glycosides or certain anti-arrhythmic drugs.

Therapy

No specific information is available on the treatment of overdose with TELCORD H. The patient should be closely monitored, and the treatment should be symptomatic and supportive depending on the time since ingestion and the severity of the symptoms. Serum electrolytes and creatinine should be monitored frequently. If hypotension occurs, the patient should be placed in a supine position, with salt and volume replacements given quickly. Telmisartan is not removed by haemodialysis. The degree to which hydrochlorothiazide is removed by haemodialysis has not been established.

Ingredients

Active ingredient

Telmisartan
Hydrochlorothiazide

Inactive ingredient

Mannitol
Sodium Hydroxide pellets
Meglumine
Povidone K-30
Purified water
Magnesium Stearate
Sodium Stearyl Fumarate
Lactose Monohydrate
Microcrystalline Cellulose
Ferric Oxide red

Storage Conditions

Do not store above 30°C.
Store in the original package in order to protect from moisture.

Dosage forms and packaging available

Telmisartan and Hydrochlorothiazide Tablets are available in Alu-Alu blister pack of 10 Tablets. Each printed carton contains 3 such blisters. (3 X 10 Tablets)

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